

LXTUL: measured results

MORPH, one 5090, d 1024, seq 1024, 480 fixed validation rows · one seed per arm · as of 2026-10-07

	CE gap to plain at depth 6; lower is better	K1–K6 what loop depth earns	sampled repetition seq_rep_4; lower repeats less
plain looped model no pointer	 CE 4.0806 (the ruler)	—	 0.215
LXTUL	 CE 4.3529 +0.272	 +0.0215	 0.041 (degenerate text)
LXTUL + pointer 5000 steps	 CE 3.9280 −0.150 [−0.169, −0.132]	 +0.0155	 0.307
+ DITTO phase 1000 more steps	trainer val +0.071 vs its control phase not measured on fresh rows	 +0.0294 †	 0.140 (control phase 0.291)
			real text 0.020

K1–K6 = CE at loop depth 1 minus CE at depth 6 · whisker = 95% CI · seq_rep_4 = 4-gram repetition of samples at temperature 0.7, top-k 40

† measured on rows the DITTO phase re-trained · the plain ruler has no pointer head; a MORPH plain + pointer ruler is owed

lab/experiments/mixed/2026-10-06-lxtul-pointer.md lab/experiments/failures/2026-10-07-lxtul-pointer-coverage.md

lab/experiments/mixed/2026-10-07-lxtul-pointer-ditto.md .agents/notes/implemented/architecture/2026-10-04-lxtul-primary-candidate.md